

ASHWIN SATISH

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EDUCATION

Northeastern University - Khoury College of Computer Science Boston, MA <i>Master's of Science, Data Science</i> Machine Learning, NLP, MLOps, Algorithms, Data Mining, Data Science Capstone	Sept 2024 - Dec 2026 GPA: 3.83/4
Christ University Bangalore, India <i>Bachelor's of Science, Computer Science</i>	June 2020 - May 2023 GPA: 3.53/4

EXPERIENCE

Gyori Lab for Computational Biomedicine Machine Learning Intern	Jan 2026 - Present Boston, MA
- Parsed 35M PubMed records, identifying 278K+ publications linked to registered clinical trials & built TrialSynth's end-to-end clinical trial LLM extraction pipeline, turning 13K+ publications into structured trial datasets for knowledge graph integration. - Designed & implemented CoGEx's Trial Result schema, converting trial datasets to 7 Neo4j node types & 9 relationship types, generating 115K+ nodes & 116K+ relationships, enabling queryable clinical trials within the knowledge graph. - Boosted adverse event entity coverage from 32.5% to 66.8% across 13k+ clinical trial papers by building a multi-ontology extraction pipeline with systematic noise filtering. - Delivered a production Clinical Trial Browser (discovery.indra.bio) with 13 REST APIs & a Flask search interface, enabling real-time lookup of trial outcomes across the CoGEx biomedical knowledge graph.	
Northeastern University Machine Learning Researcher	Jan 2025 - Present Boston, MA
- Boosted true positives from 2% to 77% while maintaining zero false positives for early-stage Lyme cell detection via 7-model CNN ensemble (TensorFlow). - Built a pipeline for reconstruction & masking of 10,000+ cell images, saving the biology team weeks of manual work. - Improved accuracy by 13% via augmentation & L2/dropout regularization, achieving production-ready performance.	
Northeastern University Teaching Assistant (Foundations of Data Science)	Aug 2025 - Dec 2025 Boston, MA
- Mentored 300+ students in linear algebra & statistics for ML & Data Science via problem-solving sessions & NumPy demos. - Conducted 4+ weekly office hours clarifying topics like constraint optimization, low-rank approximation, & regression.	
CarerAI Data Science Intern	Jul 2023 - May 2024 Gurgaon, India
- Prevented 2,000+ unnecessary procedures for customers by uncovering 500+ surgery avoidance cases through KNN analysis of patient records against historical data. - Leveraged EHR for a risk-score, improving classification accuracy by 26.4%, increasing confidence in surgery avoidance cases. - Improved KNN's accuracy by 20% using DBSCAN to remove record outliers, reducing false positives in surgery recommendations.	

SKILLS

Programming | **Data:** Python, SQL, R, Pandas, NumPy, PySpark, Scikit-learn, TensorFlow, PyTorch, Hugging Face, OpenCV
Visualization | **Cloud:** Git, Tableau, Matplotlib, Seaborn, AWS, GCP, Azure, PostgreSQL, Snowflake, Docker, Neo4j
AI | **Modeling:** Transformers, BERT, LoRA, AutoML, T5, LangChain, LangGraph, Kubeflow, Ollama, PydanticAI, MLFlow

PROJECTS

AI-Powered Pokémon Battle Strategy Engine | [Github](#)

- Built & dockerized hybrid **Multi-Agent System (MAS)** integrating **Llama 3.2** with deterministic backend for strategy analysis with a recommendation system for optimal team synergies by ingesting **1M+ aggregated usage stats** via an ETL pipeline.
- Engineered a RAG pipeline via ChromaDB & Sentence-Transformers, **indexing 40,000+ strategy docs** to ground LLM inference in semantic context.

PantryPilot (2nd Place | MLOps Innovation Expo @ Google) | [Github](#)

- Contributed to the fine tuning, evaluation & automated deployment (**GCP**) of **Llama 3.2 3B** for personalized inventory recipes, resulting in **50% reduction** in **dietary violations** via LoRA adaptation.
- Collaborated on DPO feedback with Teacher model (**Llama 3 70B**) for preference labelling, VLM for OCR & Veo 3.1 for video.